

Problem 1 - e

From the results:

Part (b) INT8 has file size of 5.74 KB and accuracy of 1

Part (b) float16 has file size of 8.95 KB and accuracy of 0.98

Part (b) dynamic has file size of 8.17 KB and accuracy of 0.98

Part (c) pruning has file size of 14.14 KB and accuracy of 0.963

Part (d) knowledge distillation has file size of 6.10 KB and accuracy of 0.96

So INT8 quantization has the smallest model size and the best accuracy.

One possible strategy to further improve the performance is to apply pruning to the student model in part d and strip it, then finally convert it to INT8 TFLite.

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	18
1	0.95	0.95	0.95	21
2	0.93	0.93	0.93	15
accuracy			0.96	54
macro avg	0.96	0.96	0.96	54
weighted avg	0.96	0.96	0.96	54

	Model	Model Size (KB)	Test Accuracy
0	Part (b) INT8	5.740	1.000000
1	Student + Pruning + INT8	3.625	0.962963

After the process, the model size further decreases to 3.625 KB, while still maintaining a good accuracy of 0.962963. This works because I set the final sparsity to be very low, so it preserves enough capacity for the pruning process. Then the final accuracy remains high.

Problem 2 - (1):

The free-tier Edge Impulse has those learning blocks:

Classification (Supervised) - One example model can be from motion data, which classifies idling, walking, running, or jumping.

Regression (Supervised) - One example model can be predicting a continuous value, such as estimating temperature from a sensor.

Anomaly Detection (GMM) (Unsupervised) - It uses a Gaussian mixture model (GMM) that finds outliers in new data, which can be used to detect unseen gestures from motion data.

Anomaly Detection (K-means) (Unsupervised) - It detects unknown states such as unexpected sensor readings like vibration data, which groups the data points that are similar into a predefined number K of clusters.

Anomaly Detection (Unsupervised) - It finds outliers in the data, for example, it will detect unusual vibration signals in a set of normal vibration data.

(2):

Dense - It is a fully connected layer, which is the simplest form of a neural network layer. The layer is used for processed data.

1D Convolution / pooling - The layer learns features that take spatial information into account along a single dimension. It is used for raw data.

2D Convolution / pooling - It learns features that take spatial information into account along two dimensions. It is used for raw data.

Reshape - It turns one-dimensional data from a DSP block into multi-dimensional data. It is used for deep learning on raw data, or to process MFCC output.

Flatten - The layer flattens multi-dimensional data into a single dimension.

Dropout - It reduces the risk of a model overfitting your dataset by randomly cutting a fraction of network connections during training.

(3):

Quantized (int8) and Unoptimized (float32)

(4):

The free-tier Edge Impulse synthetic data generation mainly supports image data and audio data. For images, available methods include Black Forest Labs Flux-Pro Synthetic Image Generator and DALL-E 3 Synthetic Image Generator. For audio, available methods include Whisper Synthetic Voice Generator.

Synthetic data is important because it helps increase the size and diversity of a dataset when real data is hard to collect. It can fill missing cases and improve robustness in the machine learning applications.